

Self-similar solutions to the time-fractional Porous-Medium Equation

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Abstract

We consider the time-fractional porous-medium equation

$$\partial_t^\alpha u = \Delta(u^m), \quad 0 < \alpha < 1,$$

in $\mathbb{R}^d$, with a Dirac mass as initial datum. The presence of the Caputo time derivative introduces memory into the evolution and modifies several familiar features of the classical porous-medium equation. In this talk, we describe a family of mass-preserving self-similar solutions and the qualitative properties of their profiles.

We prove existence in every dimension $d \geq 1$ precisely in the optimal range

$$m > m_c := \frac{(d-2)_+}{d}.$$

After reducing the problem to a nonlinear integral equation for the self-similar profile, we use sub- and supersolution methods to treat the different diffusion regimes. For $m > 1$, the profiles are compactly supported, and we determine their sharp behavior at the free boundary. For $m_c < m < 1$, the profiles are strictly positive and have algebraic tails forming an asymptotic family around an explicit very singular solution. We also describe the singular behavior near the origin, where the effect of the initial Dirac mass remains visible in the profile.

These results provide a time-fractional counterpart of the classical Barenblatt theory and illustrate the interaction between nonlinear diffusion and temporal memory.

This work is based on:

D. Gómez-Castro, Ł. Płociniczak, and J. L. Vázquez, “Self-similar solutions to the time-fractional Porous-Medium Equation,” arXiv:2604.09281 [math.AP], 2026.